

Airbnb Investment Analytics ETL & BI Dashboard

A Tableau Prep and Tableau Desktop project for Airbnb investment analysis, occupancy-rate evaluation, ROI comparison, and BI decision support in Sicily.

Project by Davoud Kalantari

Tools & Stack

Tableau Prep | Tableau Desktop | Excel | CSV | ETL | BI Analytics

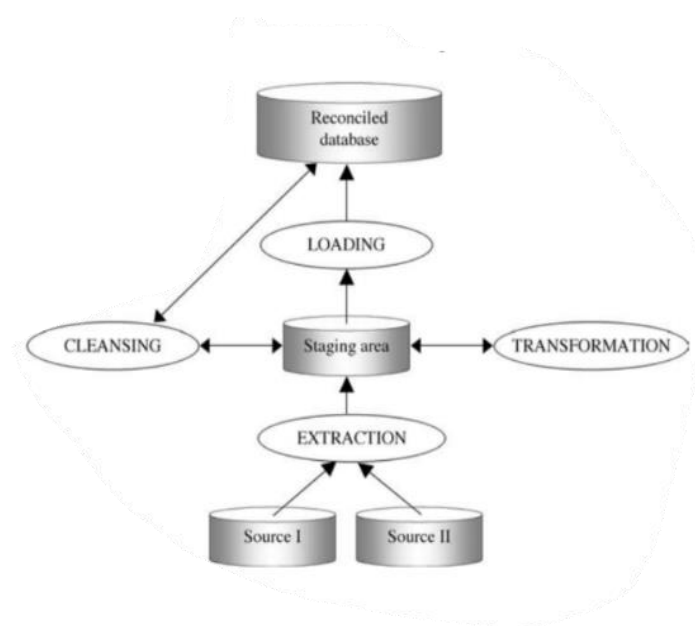
This public edition replaces the original academic cover page with a professional portfolio cover while preserving the technical body of the report.

Public portfolio edition

Description of Dataset and definition of Facts, Dimensions, and Measures for the Airbnb Examples, Output file and ETL operator

FOREWORD: This report aims to recount every detail of the project assigned during the semester but not only; the objective I would like to give is to also give a logical and not only mechanical explanation of what was carried out, which is why I personally decided to combine the ETL explanation with what was carried out in the "TABLEAU DESKTOP" software, based on the slides seen and discussed in class. Have a good look.

ETL, which stands for Extraction, Transformation and Loading, is a process used to integrate and transform data from different sources into a single repository, usually a data warehouse. Each stage of the ETL process is critical to ensure that the data is accurate, consistent and ready for analysis. Let us look at each stage in detail.



The Extraction phase consists of retrieving data from the various original sources. These sources can be relational databases, flat files, web applications, XML files, APIs, and more. The aim is to collect all the necessary data in a raw, unprocessed form.

Sub-phases of Extraction:

1. **Identification of Data Sources:** Determine which data sources are relevant to the analysis.
2. **Connection to Data Sources:** Establish secure and reliable connections to data sources.
3. **Data Retrieval:** Query or read data from the sources.
4. **Validation and Verification:** Ensure that extracted data is complete and accurate by checking data integrity.

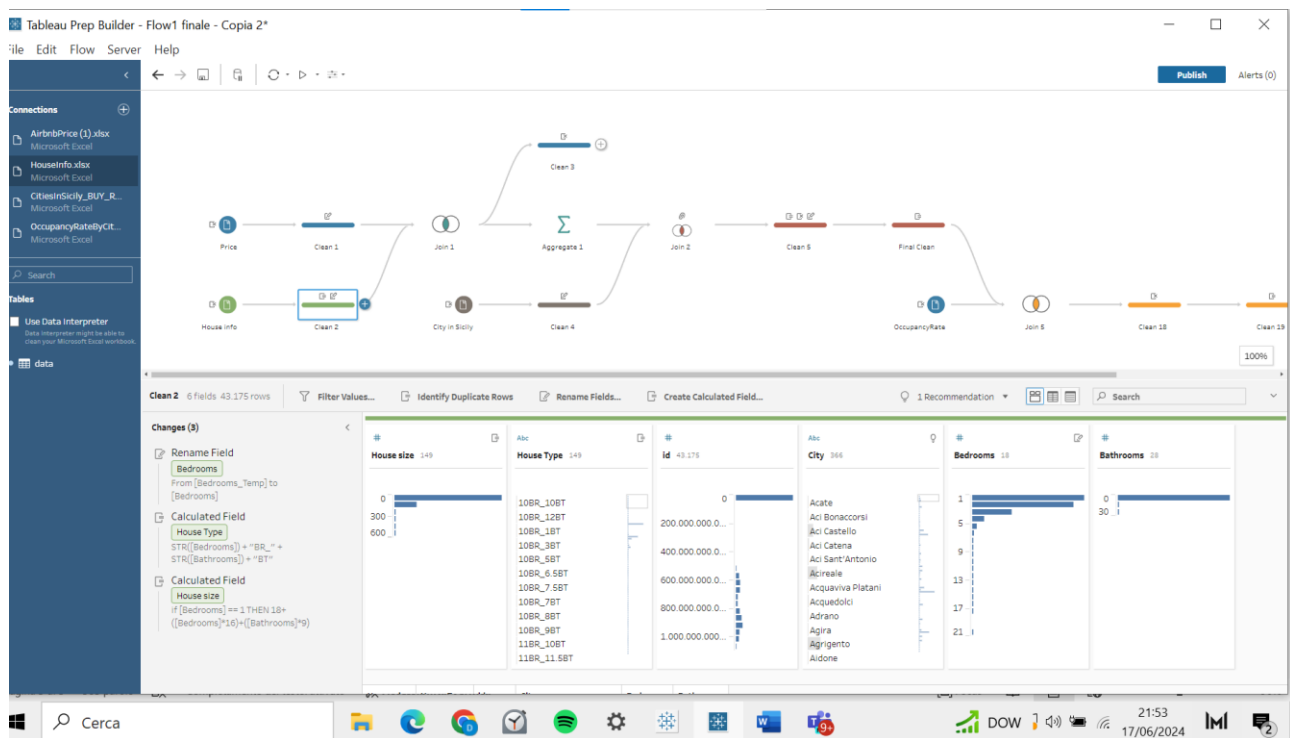
On the basis of what was mentioned and seen during the lessons held in class, this initial phase can be recounted with the collection of 3 excel files and the following one: these files are initially **"Airbnb price"**, **"House info"**, **"City in Sicily by rent"**, of which a fourth excel file will be provided later on, named as **"Occupancy rate by city"**. Briefly we can give an explanation of what these files contain: the first one, "Airbnb price", relates the prices for each id code, the second one, "House info" gives us information, given the respective id, on the cities in Sicily and on the "bedroom temp" and "bathrooms" which tells us for each house how many bathrooms and lounges they have. "City in Sicy by rent", on the other hand, relates the Sicilian towns contained in the excel file used for the price of their sale but also their rent. Finally, we introduced (but only later) a fourth file named "Occupancy rate by city": this file, which will be fundamental for subsequent analyses, allows us to see what are the occupancy rates calculated correctly and no longer based on a previous estimate. for each municipality contributed. This step just told by me is fundamental in recognising the first sentence of the three of the ETL.

TRASFORMATION:

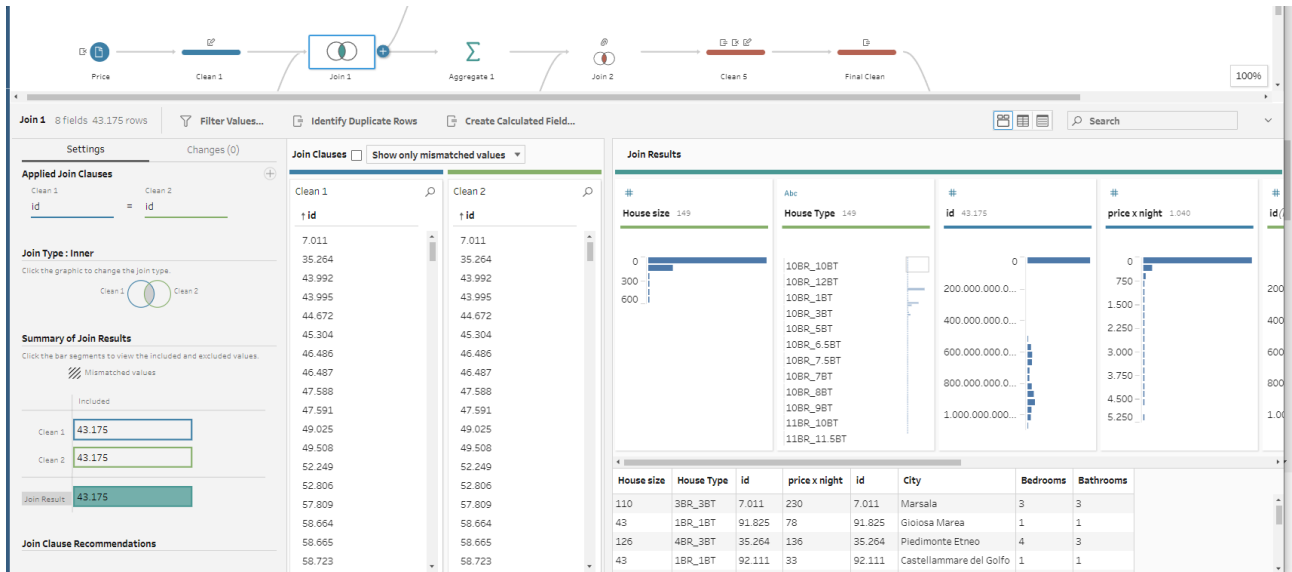
Once the data has been extracted, it must be transformed to suit the needs of the data warehouse and end users. The transformation includes a series of operations to clean, aggregate and convert the data into the desired format. This phase is also divided into sub-phases:

- **Data Cleaning:** Correct or remove incorrect or incomplete data (e.g. removal of duplicates, correction of typing errors).
- **Standardisation:** Standardise data into a common format (e.g. convert all dates into YYYY-MM-DD format).

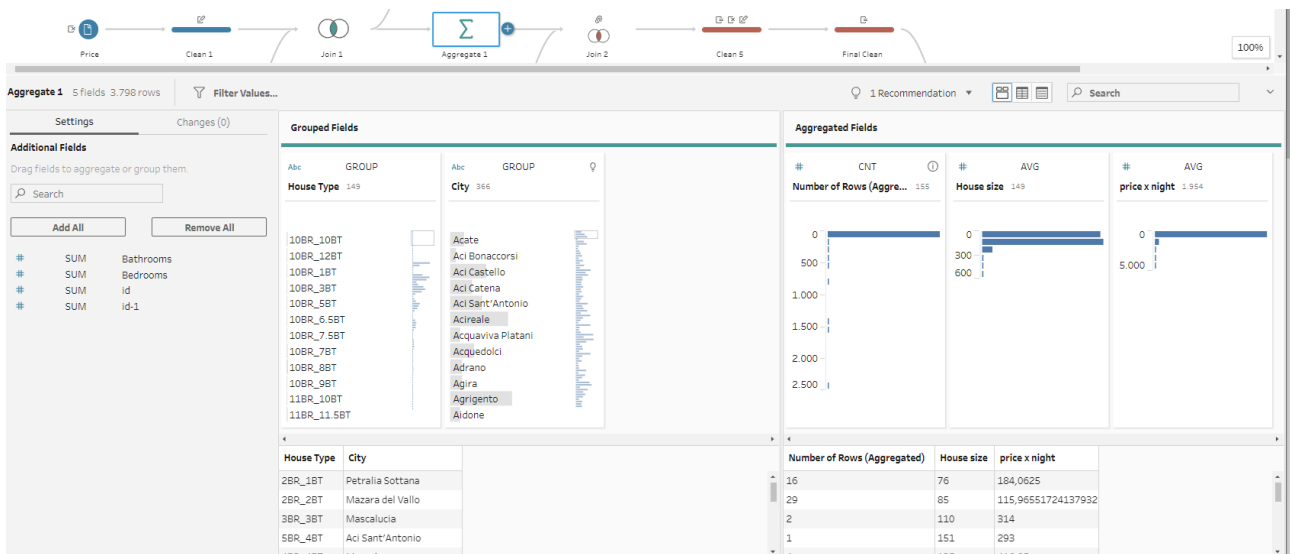
- **Aggregation:** Summarise data according to different dimensions (e.g. sums, averages, counts).
- **Integration:** Combine data from different sources to create a unified view



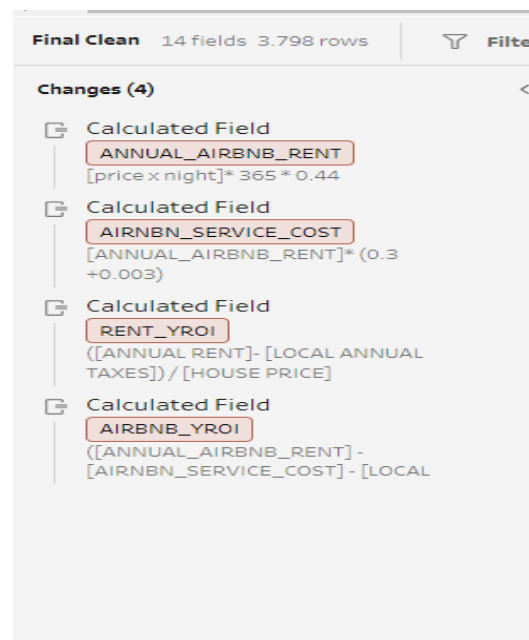
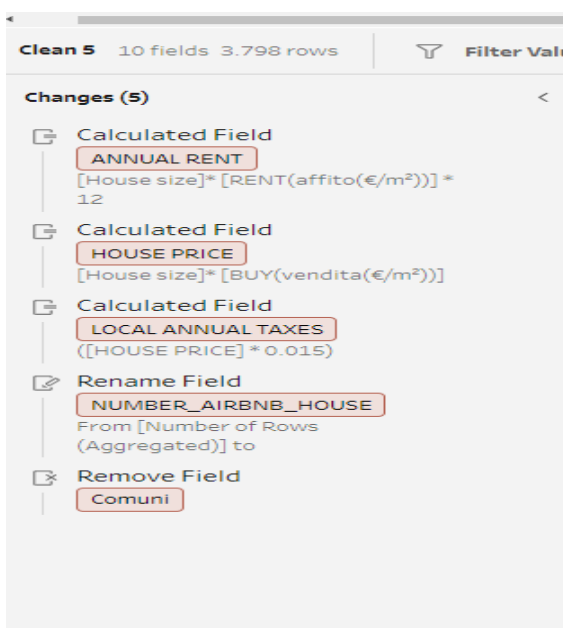
Now let's see specifically what I did: After linking the 3 excel files in the software, I proceeded to make some "cleans" in order to fix and improve some categories of data. Specifically, in "clean 1" I changed the name from price to "price x night", while in clean 2 I changed the name from bedroom_temp to "bedroom" and then I created two calculated fields: "Housetype" (gives me the name of the house types) and "housesize" (gives me its dimensions) by means of calculation operations which you can see in the prep. The next part concerns the JOIN (JOIN 1). The JOIN is part of the transformation in that it allows rows from two tables to be joined (in this case I am joining the two cleans) via a common value (ID)



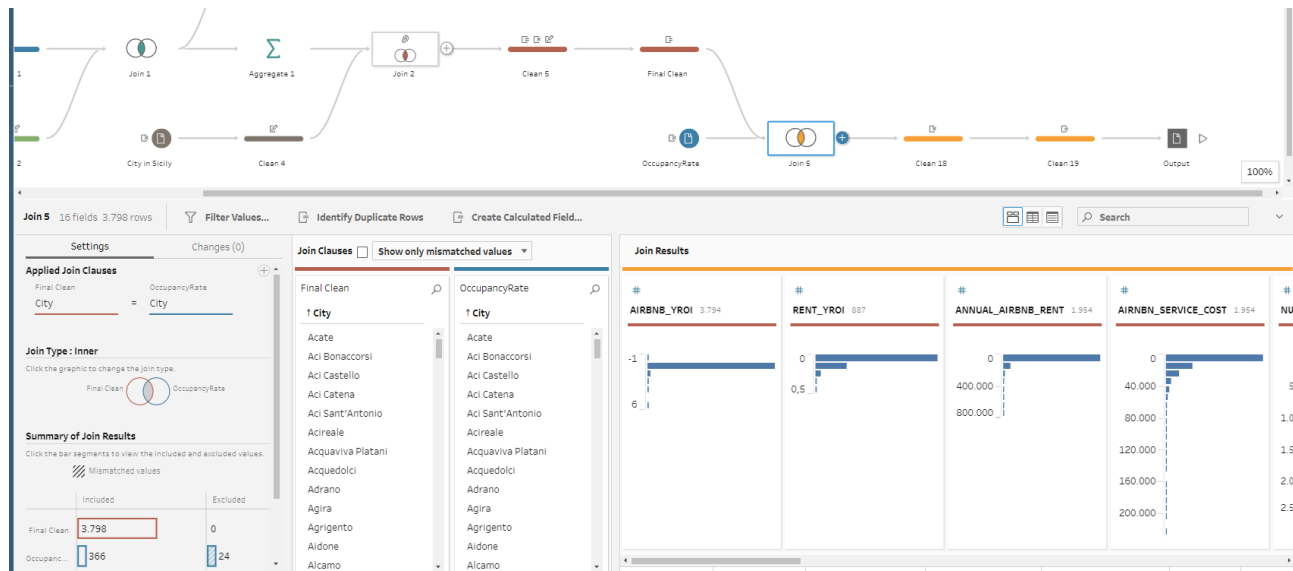
After doing the join and also a 'clean 3' linked to it to remove the 'id-1' field that was completely useless for our purpose, I continued the journey with '**AGGREGATION**'. Aggregation is usually used to summarise data by means of calculations such as sums, averages or counts.



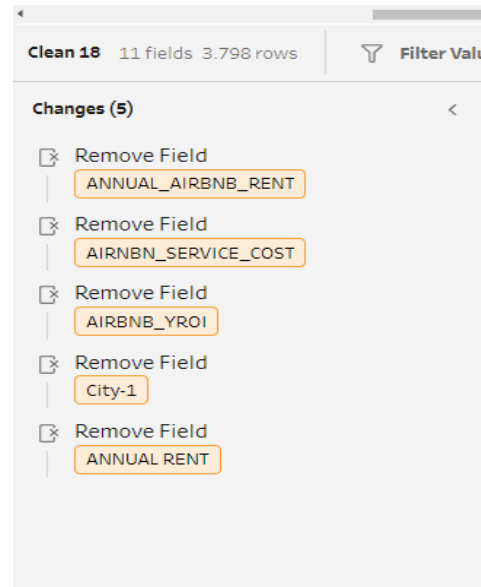
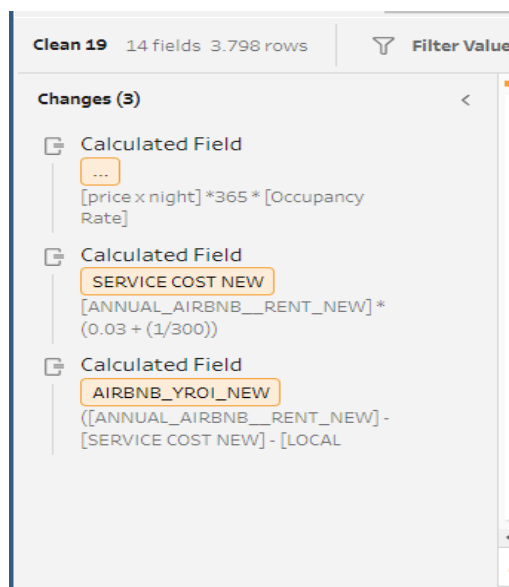
In this case we aggregated 5 fields of which housetype and city are grouped by grouped field while I decided to put number of rows as COUNT and "house size" and "price x night" as AVERAGE. In parallel with this operation, I inserted the third file exceò "City in Sicily" in the path and after doing another clean in which I renamed the sale and rent, I did a second JOIN, named as JOIN 2 between the two and then did the "clean 5" and the "final clean"



The most important operations in these two phases were to calculate: the local annual taxes, house price the number of airbnb and remove the municipalities, while in the last clean we calculated fields that will be fundamental in the following phases such as the return on investment of the airbnb or the return on investment should we decide to put the houses up for rent, but also the cost of the service that will be a fundamental field for the later calculation of the adjustment of our airbnb yroi.



Finally, we added the last excel file of our prep "Occupancy Rate" and did the last join (join 5) between the final clean and this excel file, and then did the last two cleans (clean 18 and clean 19)



LOADING:

The last step in the ETL process **is the loading of the transformed data into the data warehouse.** This step must be performed with care to minimise the impact on the data operations performed before moving on to the visualisation of the transformed data.

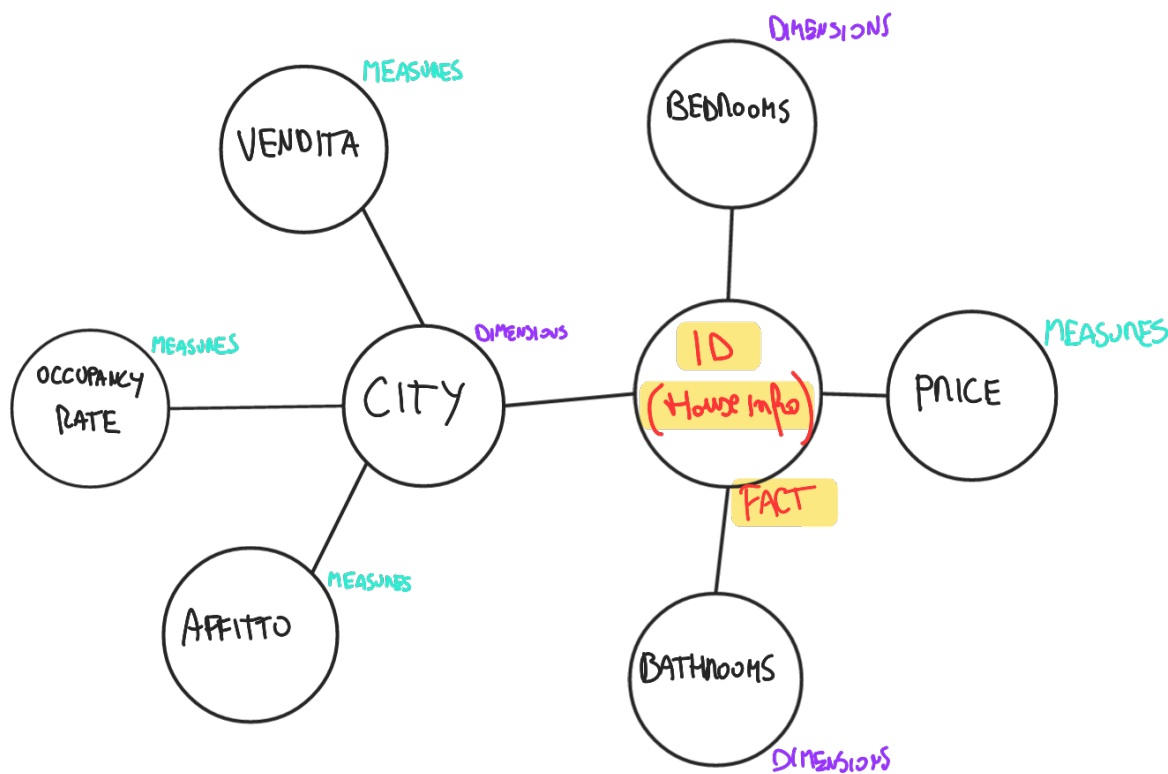
However, Tableau Prep does not perform an actual loading into a traditional data warehouse. Instead, the end product of Tableau Prep is usually a data stream that is used directly in Tableau for visualisation and analysis. that if Tableau Prep does not load the data into a data warehouse, the output can be seen as the equivalent of 'loading' in the context of direct use in Tableau. This is because prepared data is ready for analysis, just as data loaded into a data warehouse is ready for queries and reports.

At the end of our prep, our **output** will thus consist of:

- AIRBNB_YROI_NEW
- ANNUAL_AIRBNB_RENT_NEW
- SERVICE_COST_NEW
- RENT_YROI
- NUMBER AIRBNB
- HOUSE PRICE
- LOCA_ ANNUAL_ TAXES
- HOUSE SIZE
- HOUSE TYPE
- CITY
- PRICE X NIGHT
- BUY
- RENT
- OCCUPANCY RENT

Now it is time to save our output for export to the desktop tableau.

Definition of Facts, Dimensions, and Measures for the Airbnb Examples: •
Provide a conceptual model and a logical design. Refer to the specific slides from the lecture on data modeling. Note that this point is required but not evaluated



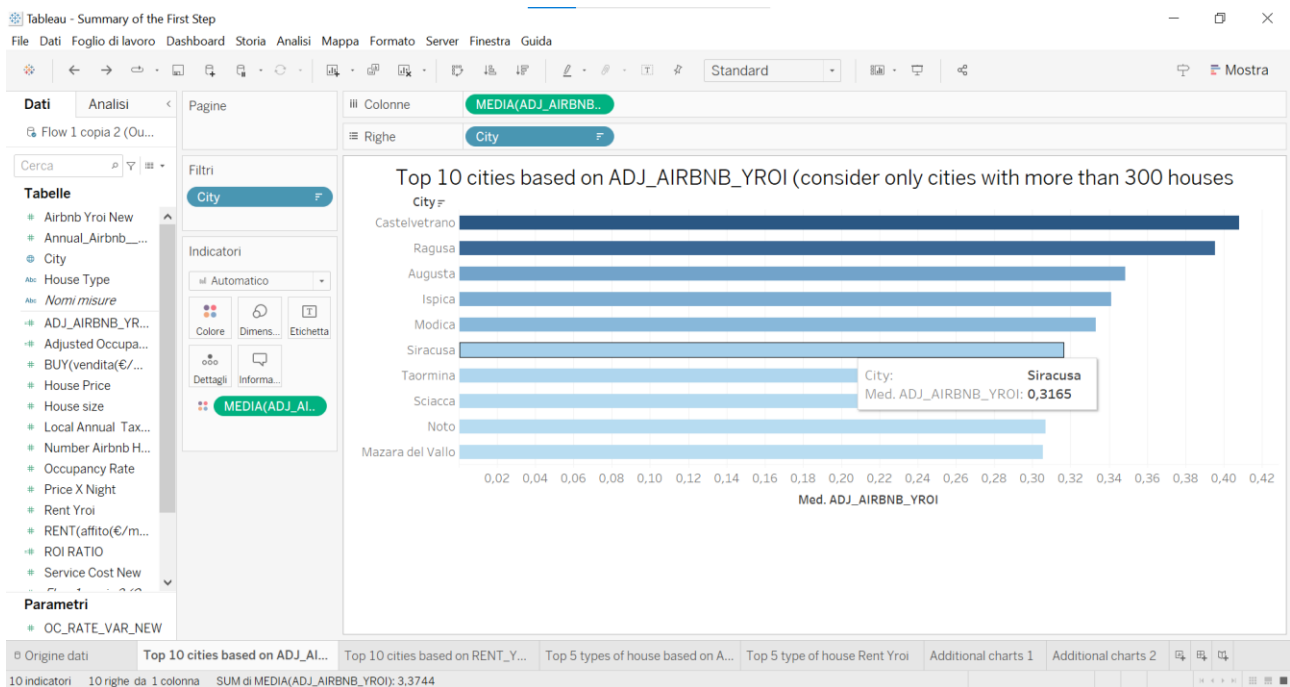
FACT SCHEMA

Summary of the First Steps:

NB: For these graphs, I used the final prep output. I also decided after step 4 to use as the "occupancy rate" the so called "Adjusted occupancy rate", i.e. the occupancy rate given by the excel file you reported, adjusted by the new interest rate given in step 4 which fluctuates from -5% to 5%.

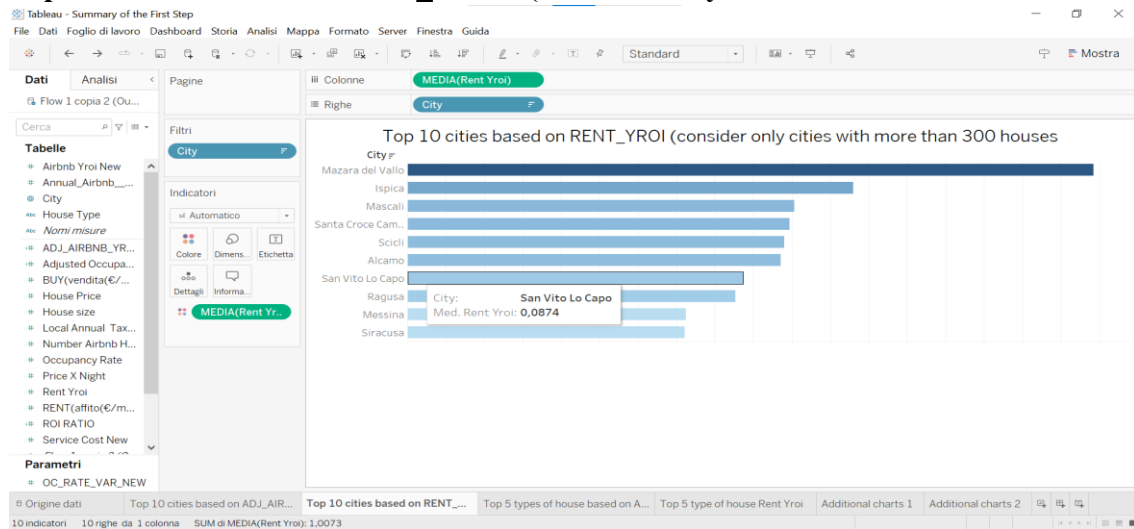
I therefore manually added the missing values such as ADJ_AIRBNB_YROI which I had not put in the output file but created in step 4 of tableau desktop. The reference values are from the last step

- **Top 10 cities based on ADJ_AIRBNB_YROI (consider only cities with more than 300 houses)**

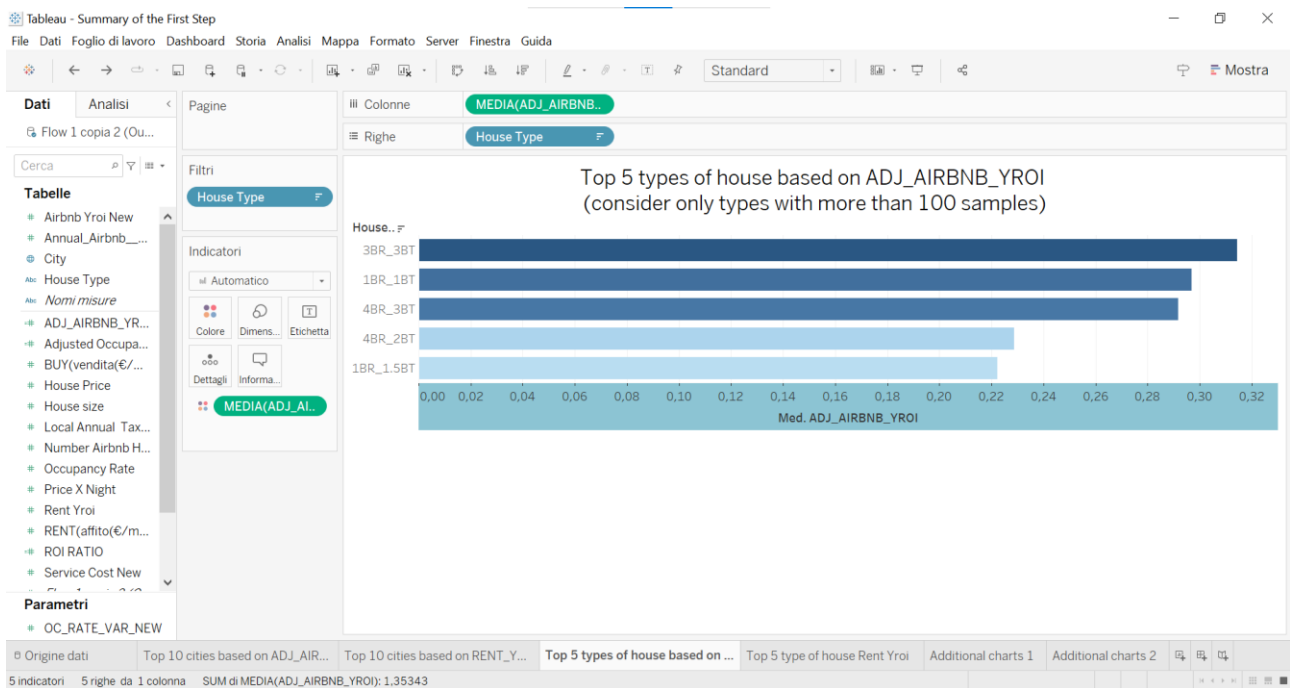


In this graph as in the next three graphs, I decided to use the bar graph and to use a sequential blue palette. The reason for this stems from the fact that the colour used as a filter indicates in this case the average of ADJ_AIRBNB_YROI, with colour gradation to indicate to me an increase of the same value. In this traffic and the next two, I decided to sort the cities in a descending and "nested" manner by the reference value (ADJ_AIRBNB_YROI in this case) to make the graph more orderly.

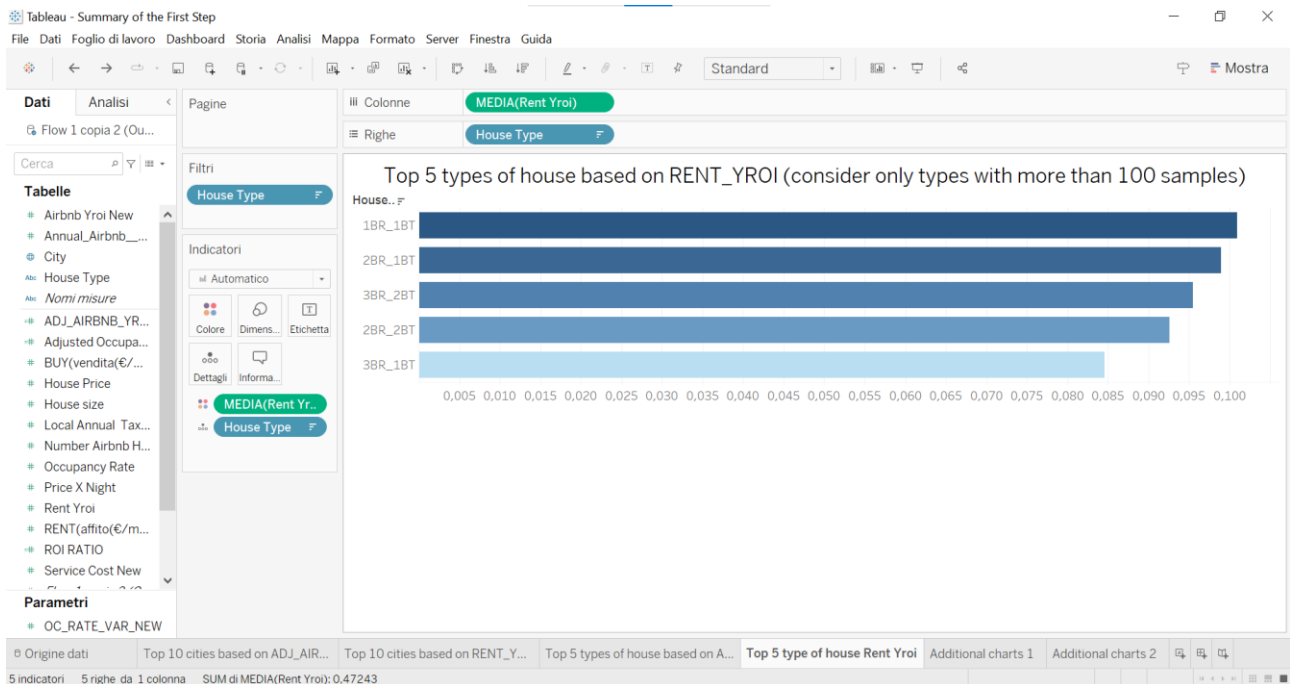
- **Top 10 cities based on RENT_YROI (consider only cities with more than 300 houses)**



- **Top 5 types of house based on ADJ_AIRBNB_YROI (consider only types with more than 100 samples)**

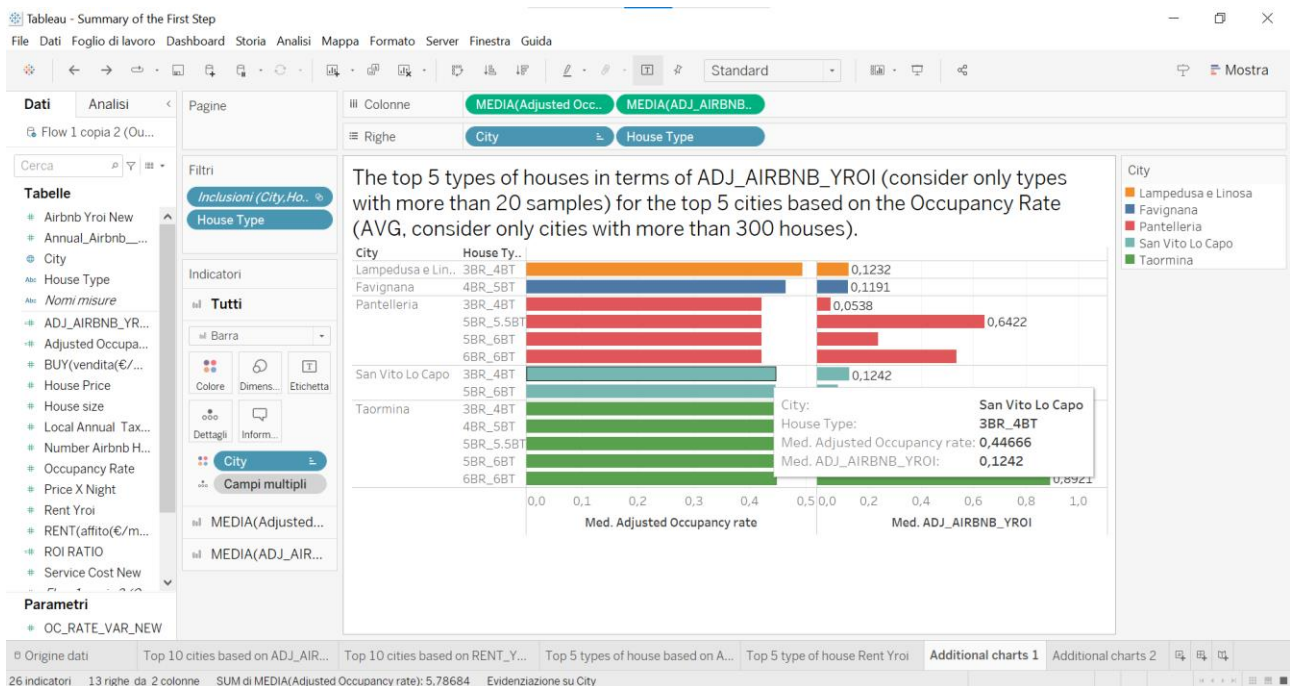


- **Top 5 types of house based on RENT_YROI (consider only types with more than 100 samples)**

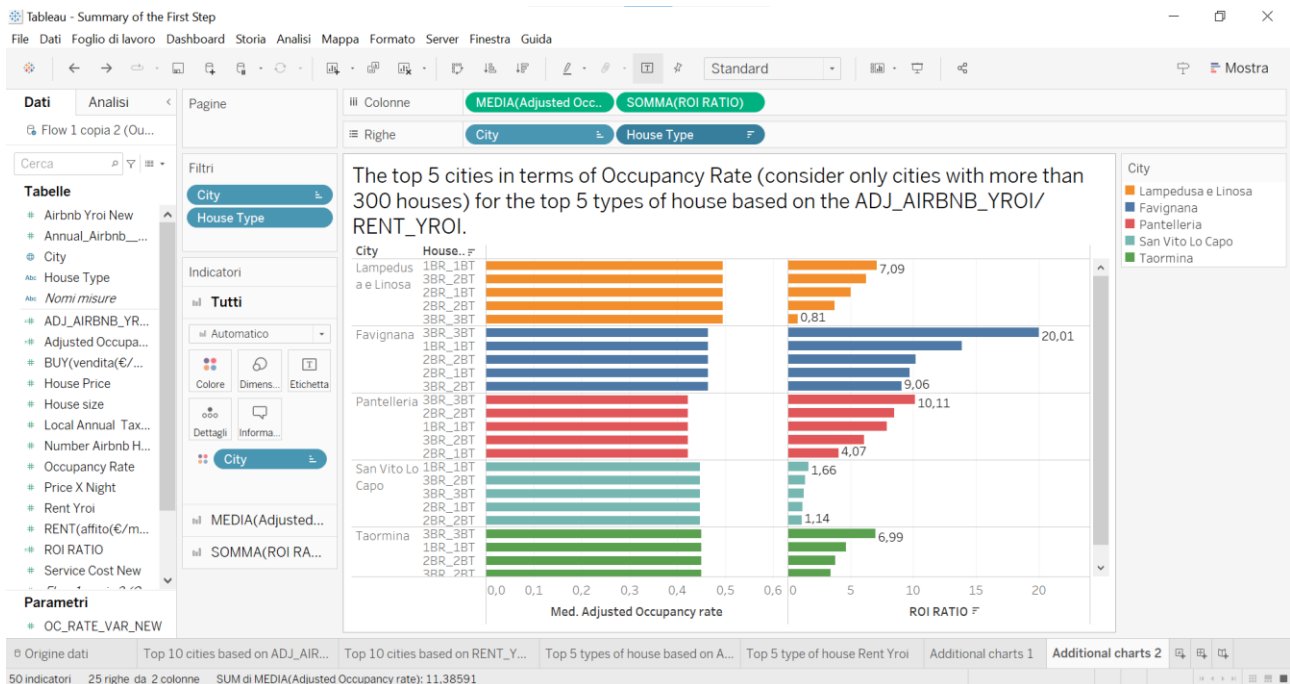


Additional Summary Charts:

- **The top 5 types of houses in terms of ADJ_AIRBNB_YROI (consider only types with more than 20 samples) for the top 5 cities based on the Occupancy Rate (AVG, consider only cities with more than 300 houses).**



- The top 5 cities in terms of Occupancy Rate (consider only cities with more than 300 houses) for the top 5 types of house based on the ADJ_AIRBNB_YROI/RENT_YROI.



In the latter two graphs, however, I decided to use **different colours**; for example, I created a colour filter for the cities so that it was easy to distinguish each type of city by colour, deciding to leave the map to make it immediately understandable which city and type of house it refers to.

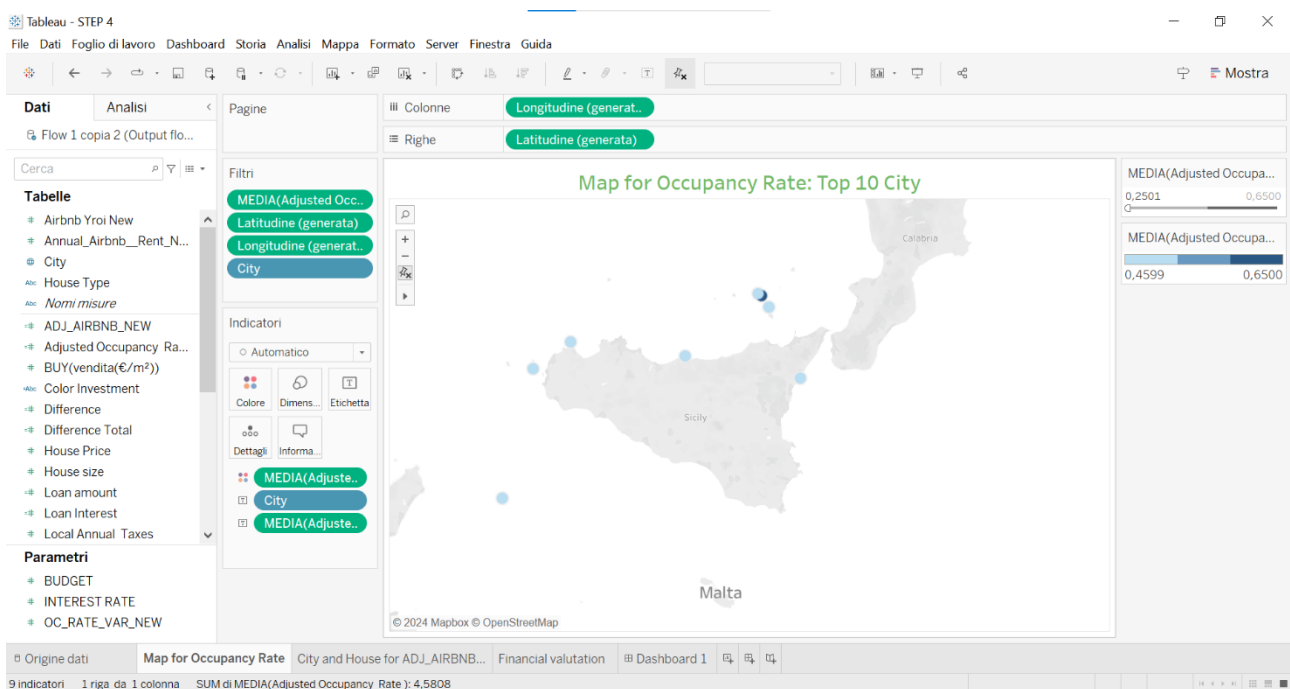
I preferred to put the labels only on the ADJ_AIRBNB_YROI (FIRST OF TWO GRAPHS, IN RELATION TO HOUSETYPE) and ROI RATIO (derived from the ratio between ADJ_AIRBNB_YROI/RENT_YROI). The decision to put the label only on the minimum value and the maximum value and not on all the values derives from the fact that according to the theory studied, since the graph is made up of many bars, it does not make sense to put the label on every value as this would make the graph difficult to read. In addition, the values are very similar to each other of the "OCCUPANCY RATE" which is why I decided to avoid placing each value as placing the label would have made the graph even fuller and less distinguishable, adding values that would result in the results being returned

Again, I preferred to sort the data in DECREASING order, to increase the general order of the graphs

STEP 4

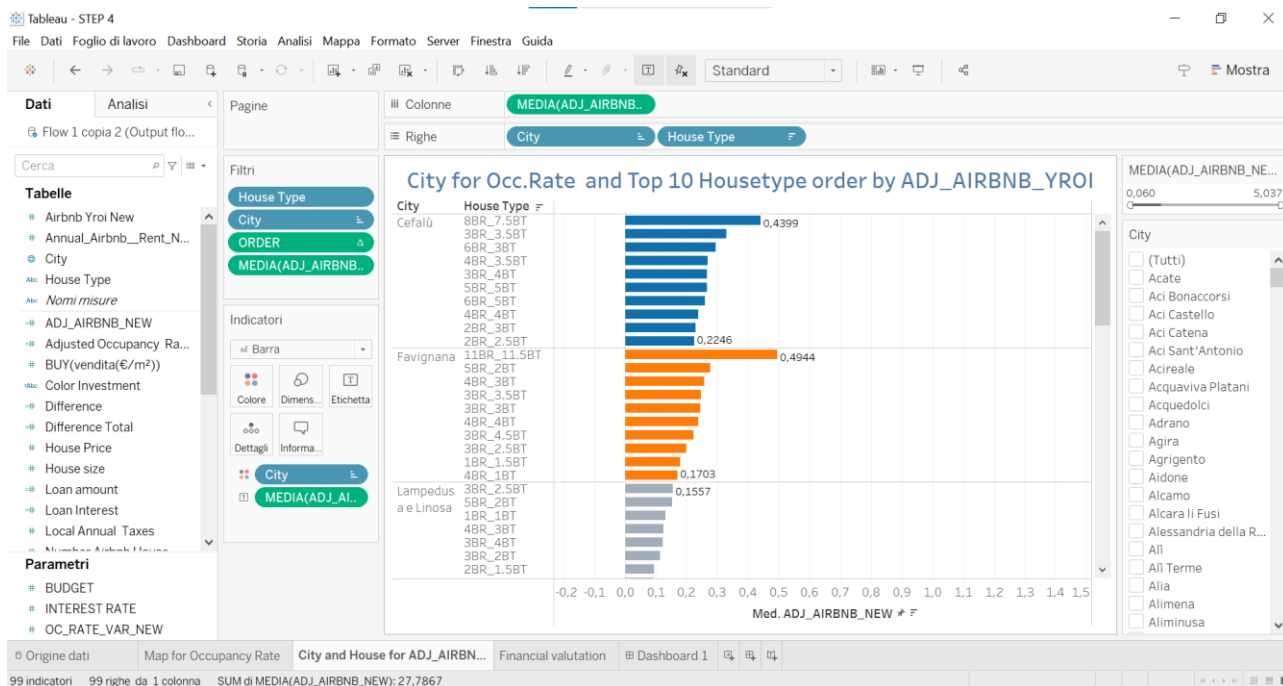
Once we had exported our final output from tableau prep to tableau desktop, it was time for the realisation of our exploratory dashboard. First of all, I adjusted and added a few values: For example, I created a new "OC_RATE_VAR" as a parameter that fluctuates from -5% to 5%, and I chose to make the new "ADJ_AIRBNB_YROI" directly on the desktop tableau, taking into account the new fields calculated in the final stages of prep (AIRBNB YROI and SERVICE COST) by replacing this OC_RATE_VAR. Afterwards, in the realisation of the dashboard, I proceeded to create 3 separate worksheets:

1) Map for Occupancy Rate: Top 10 city



In this case I first modified our "Occupancy Rate" as "Adjusted occupancy rate", this is because I decided to sum up the given occupancy rate of each city and "adjust" it with the given variation to be precise. Then the first worksheet was made comparing the top 10 cities for average occupancy rate by selecting the top 10 cities with the highest occupancy rate. This factor is important because if one decides to put up b&b houses, one must take the occupancy rate into account in order to ascertain the occupancy of the population in that city or those selected cities.

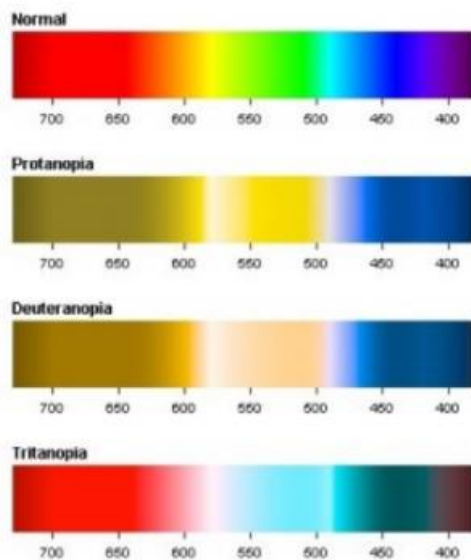
2) CITY FOR OCCUPANCY RATE and TOP 10 HOUSETYPE order by ADJ_AIRBNB_YROI



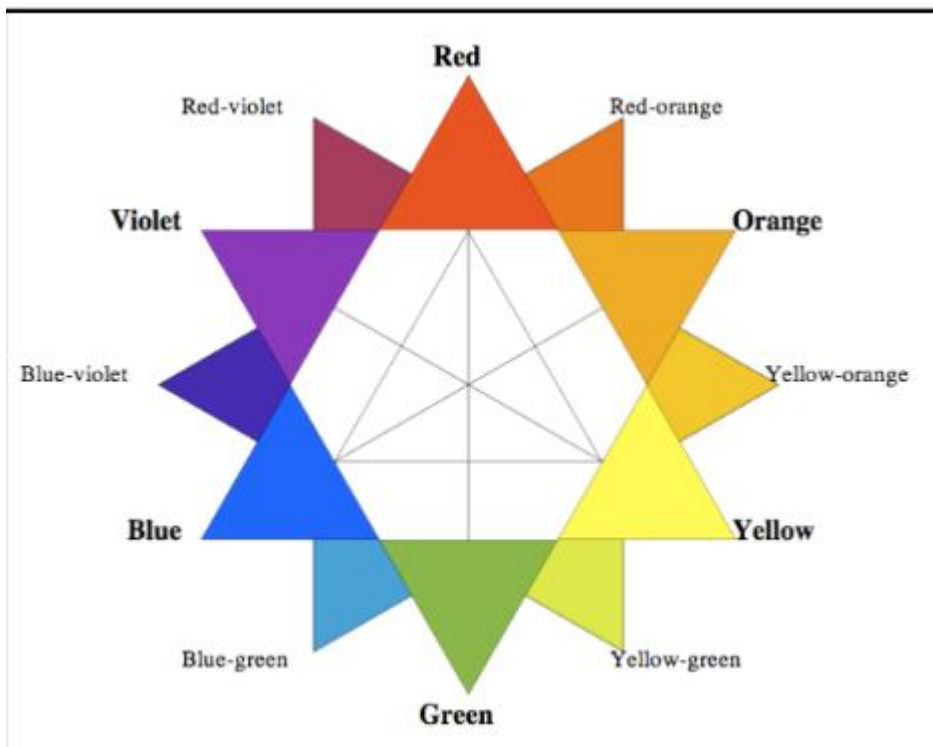
Once I have decided which are the 10 cities in which it is worth investing based on the occupancy rate, in this second tab I have decided to use the same 10 cities analysed previously and choose the 10 best types of houses that are ordered by ADJ_AIRBNB_YROI. In this case I have sorted the HOUSETYPE in descending order for each city deciding to use a label that allows us to immediately identify for each city the minimum value and the maximum value that can be obtained, with reference to the ADJ_AIRBNB_YROI. I also decided to use **another colour filter** to distinguish all 10 cities. The colour filter is not random but was chosen on the basis of the slides seen in the lecture. In this case, the goal is to create a dashboard that takes into account the entire population and, consequently, also those with visual impairments. This is why I decided to use the "colour blind" filter, as it allows us to distinguish cities clearly for anyone including those who suffer from such visual impairments.



How Color is Perceived by Someone with CVD

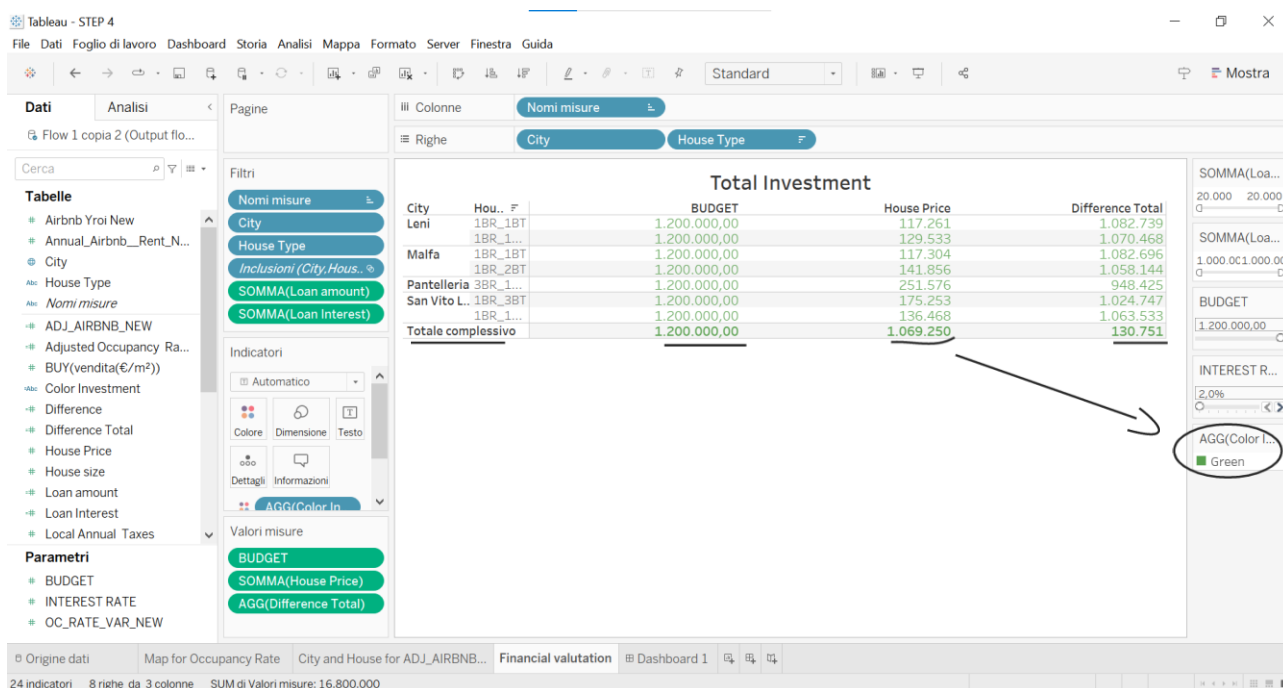


I used the color explanation file



Now with this worksheet, reusing the 10 best cities for occupancy rate and at the same time, we got the 10 best housetype for return on investment.

3) TOTAL INVESTMENT



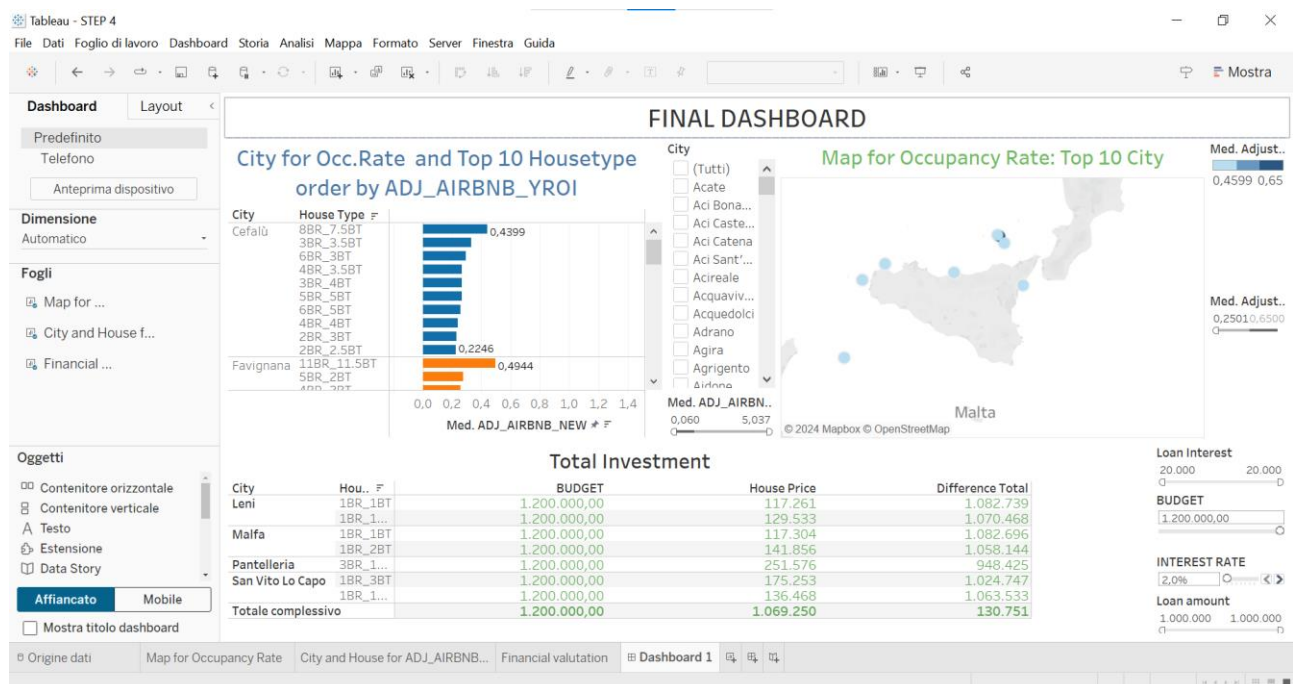
This last worksheet instead takes into account the financial sides. To do this, I decided to create a parameter called "BUDGET" which varies from 700,000 to 1,200,000 that tells us the minimum and maximum amount we can spend. Considering that 200,000 is a fixed sum that we hold from the start, it is clear that the rest of the sum we have to borrow from the bank. To do this, I decided to create another calculated field called "LOAN AMOUNT", which allows us to immediately see what the value of our budget is minus the money we hold from the start: in practice it shows us what the value of the money we borrow from the bank is. However, we also have to consider the interest rate on that loan, which ranges from 2 % to 2.6 %, so not knowing exactly the value of the interest rate, I thought of creating an interactive bar that ranges from 2 % to 2.6 % so that the customer can then make his or her own specific assessments. However, I thought it best to make such a bar **interactive**, so that if you move this bar, it will show exactly what the monetary value of the interest rate applied to the loan is, so that our hypothetical customer can see IMMEDIATELY what the amount of simple and annual interest to be paid on the borrowed money is.

Once these calculated fields were created, in this worksheet I thought it best to create a table in which I show what are the 4 best cities and the 7 best houses in the four cities as the best advice to give to our 'hypothetical client', so that he can visualise his budget, taking into account that it obviously does not exceed 1,200,000, taking into account what have been previous calculated occupancy rates and return on investment: in short, I illustrate an advice based on the analysis made in the previous sheets.

In the table I have inserted "HOUSE PRICE" as a sum that first shows us the price of each house and in the last row (TOTAL TOTAL), what is the final sum of all the houses

We have also created a "DIFFERENCE TOTAL" column (like [Budget] - SUM(HOUSE PRICE)) that allows us to understand, the difference between the budget that I have decided to spend as I wish and that I can change interactively, for all the houses. This value is important to understand whether the final purchase will be sustainable (i.e., possible to realise) or not. How do I figure this out? By using python code that I have inserted as a filter that allows me to change colour from green to red depending on whether the budget available is sufficient to realise the purchase of the houses or not (USE OF DIFFERENT COLOURS from negative to positive values as seen and studied in the visual perception slides in class)

FINAL DASHBOARD

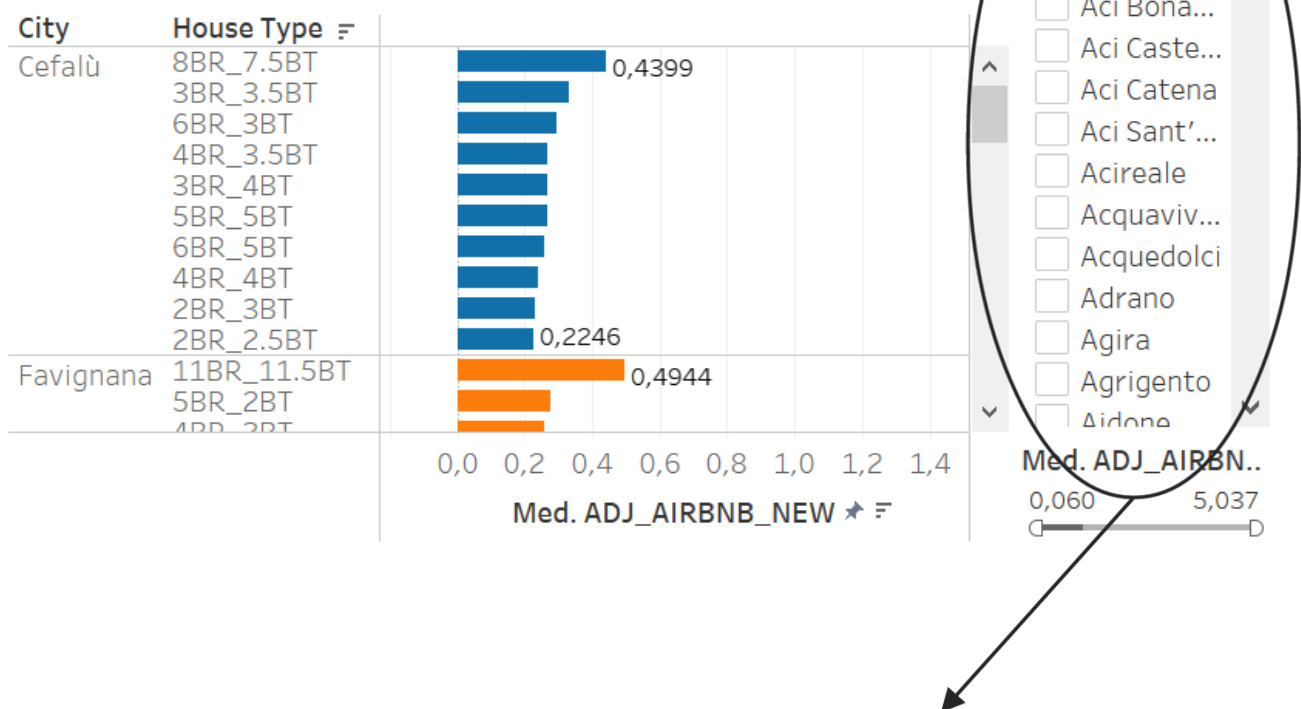


This is our final dashboard. The objective of this interactive dashboard is to create a dashboard that allows any user to be able to make their own personal evaluations, based on a path that I indicated: that is, to recommend buying houses to rent in AIRBNB, that are located in cities that fall within the best 10 cities for occupancy rate and of these 10 cities to select the best housetype for return on investment when renting in Airbnb. In the end on the final project request I therefore decided on four

cities and 7 housetypes that fall within the requirements I set out above that allow me to have a good balance between occupancy rate and return on investment while respecting the budget.

The interaction was made possible by some interactive bars inserted: for example I give the possibility of being able to view other cities that do not fall within the range indicated by me for the rate of return on investment, but I give free choice to anyone to be able to consult the return on investment of other cities should they decide to change their mind. In short, the objective of this dashboard is to give an indication, an advice, but then to be able to move city budgets or other factors in a completely personal way.

City for Occ.Rate and Top 10 Housetype order by ADJ_AIRBNB_YROI



Despite the fact that I have decided to choose the cities, however, I leave it free for my "hypothetical client" to be able to view other cities of his choice based on ADJ_AIRBNB_YROI

City	Hou..	BUDGET	House Price	Difference Total
Leni	1BR_1BT	1.051.425,71	117.261	934.165
	1BR_1...	1.051.425,71	129.533	921.893
Malfa	1BR_1BT	1.051.425,71	117.304	934.122
	1BR_2BT	1.051.425,71	141.856	909.570
Pantelleria	3BR_1...	1.051.425,71	251.576	799.850
San Vito Lo Capo	1BR_3BT	1.051.425,71	175.253	876.173
	1BR_1...	1.051.425,71	136.468	914.958
Totale complessivo		1.051.425,71	1.069.250	-17.824

Loan Interest
18.731 18.731

BUDGET
1.051.425,71

INTEREST RATE
2,2%

Loan amount
851.426 851.426

NEGATIVE

As mentioned earlier in worksheet 3, another decision I have decided to introduce is that of **colours** in this case what we can notice and who is if the Budget is less than the sum of the price of all the houses and consequently the total difference (given as the difference between Budget and SUM house price) is **NEGATIVE**, It will be marked with a red colour indicating that the investment is not sustainable because we do not determine sufficient economic resources to be able to buy those houses.

if the Budget is equal to or greater than the sum of the cost of all the houses, it will tick green as it indicates to us that we can sustain that initial cost.

City	Hou..	BUDGET	House Price	Difference Total
Leni	1BR_1BT	1.200.000,00	117.261	1.082.739
	1BR_1...	1.200.000,00	129.533	1.070.468
Malfa	1BR_1BT	1.200.000,00	117.304	1.082.696
	1BR_2BT	1.200.000,00	141.856	1.058.144
Pantelleria	3BR_1...	1.200.000,00	251.576	948.425
San Vito Lo Capo	1BR_3BT	1.200.000,00	175.253	1.024.747
	1BR_1...	1.200.000,00	136.468	1.063.533
Totale complessivo		1.200.000,00	1.069.250	130.751

Loan Interest
20.000 20.000

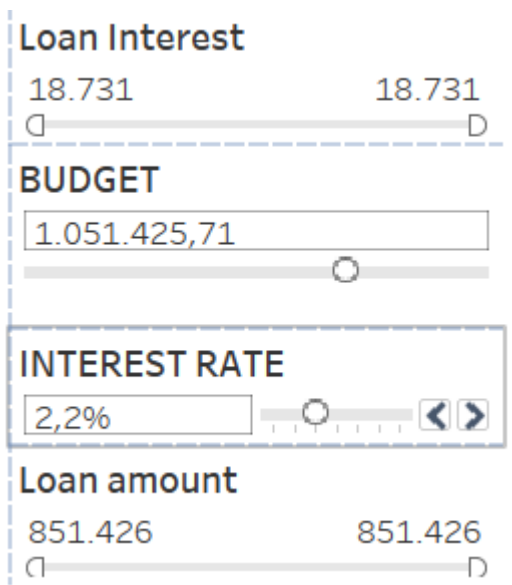
BUDGET
1.200.000,00

INTEREST RATE
2,0%

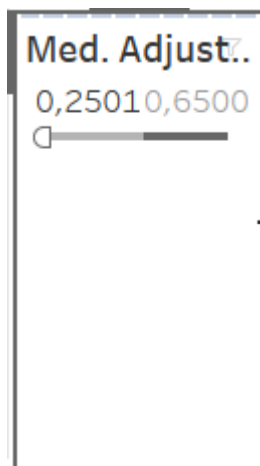
Loan amount
1.000.000 1.000.000

POSITIVE

INTERACTIVE BARS INSERTED IN THE DASHBOARD



I can maneuver my budget and interactively see how other values change



I can move my Avg ADJ_AIRBNB_YR. Or interactively to my liking.



I can maneuver my return on investment of airbnb

CONCLUSION

In conclusion; the objective was to create an interactive dashboard that on the one hand gives an analysis of the constraints you have provided but at the same time leaves room for any person to be able to make evaluations both on the cities, being able to modify them as they wish, and on the occupation rate, being able to view on the interactive map the occupation rate they prefer by moving the appropriate interactive bar and, above all, being able to enter the budget figure they prefer and at the same time understand, deciding to spend a generic value "x", (always bearing in mind that 200.000 euro is money he already has) how much he has to borrow from the bank, how much interest he will have to pay to the bank on that loan, and at the same time whether or not such a purchase will be economically feasible.